Kushal KC

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Malpi-08, Panauti, Kavrepalanchok, Nepal
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+977 9823584057
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Nepali

PROFILE

A dedicated Geomatics Engineer Student and Satellite Data Analyst with a passion for GIS Mapping and Remote Sensing. Currently pursuing a Bachelor's degree in Geomatics Engineering at Kathmandu University. While working at Antarikchya Pratisthan Nepal I was experienced in satellite data processing, spatial analysis and map creation.

EDUCATION

Kathmandu University (KU) | Dhulikhel, Kavrepalanchok, Nepal ***2025 – Expected 2029***
Bachelors of Engineering in Geomatics Engineering (B.GE)
Website: geom.ku.edu.np
Email: kushal7125@student.ku.edu.np
Key Modules: Programming (C, C++), SketchUp, AutoCad

Nepal Banepa Polytechnic Institute (NBPI) | Banepa-05, Kavrepalanchok, Nepal ***2019 – 2023***
Diploma in Geomatics Engineering (DGE)
Website: nbpi.edu.np
Aggregated Score: 77.86%
Key Modules: Surveying, Remote Sensing, Geographic Information Systems, Photogrammetry Programming (Python)

Arunodaya English Secondary School | Panauti-4, Kavrepalanchok, Nepal ***2006 – 2019***
Secondary Education Examination (SEE)
Aggregate Grade: 3.85

EXPERIENCE

Geomatics Engineering Society | Dhulikhel, Kavre, Nepal ***September 2026 - Present***
Web Designer
Website: https://ges.ku.edu.np/

Freelancer (remote) ***March 2024 – Present***
Satellite Data Analyst | GIS & WebGIS Mapping
Collaborative with **Kathmandu Metropolitan City (KMC), Ward No. 5 & Er. Sirish Pandey**

- ➢ Contributed to WebGIS-based mapping and spatial data visualization for the Development and Cultural Profile of Ward No. 5, Kathmandu Metropolitan City, including infrastructure, road networks and cultural heritage mapping.

Collaborative with **Dr. Ranadheer Mandadi**

- ➢ Contributed to Land Use/Land Cover (LULC) classification and future prediction using satellite imagery.

Collaborative with **Er. Sameer Prashad Kafle**

- ➢ Performed Land Use and Land Cover Change Detection and Impact of Road Construction: A Case Study of Pathari-Shanishchare Municipality

Collaborative with **Er. Prithivi Khadka**

- ➢ Contributed to Nakhhu Flood Modelling

Collaborative with **Deepa Khadka**

- ➢ Performed Assessment of Land Use and Land Cover Dynamics with People's Perception in Ishworpur Municipality, Sarlahi, Nepal

Antarikchya Pratisthan Nepal | Lazimpath, Kathmandu, Nepal ***August 2023 – September 2025***
Satellite Data Analyst | GIS Mapping
Website: antarikchya.org.np
Email: kushal@antarikchya.org.np
Collaborative projects with **Kathmandu Metropolitan City (KMC)**

NAXA | Maharajgunj-3, Kathmandu, Nepal ***May 2023 – June 2023***

GIS Analyst
Website: naxa.com.np
Email: kushal.naxa@gmail.com
Collaborative projects with **Changunarayan Municipality and National Disaster Risk Reduction and Management Authority (NDRRMA)**

PROJECTS & RESEARCH

Satellite Data Analytics Project (SDAP) *August 2023 – September 2025*

Satellite Data Analyst | GIS Mapping

- Executed satellite data analysis and GIS mapping for disaster and environmental applications under the Satellite Data Analytics Project (SDAP).
- Assessed flood damage in the Sapta Koshi region using remote sensing and spatial analysis techniques.
- Monitored environmental changes in the Chure region using multi-temporal satellite imagery.
- Visualized and analyzed geospatial data related to earthquakes and forest fires.
- Performed dynamic population density mapping using geospatial modeling approaches.
- Developed disaster preparedness protocols at Antarikchya Pratisthan Nepal focusing on flood and urban fire susceptibility mapping.
- Applied GIS-based multi-criteria analysis to generate susceptibility maps for disaster risk reduction.
- Designed and developed Web GIS applications using Leaflet.js, Mapbox, and MapTiler for interactive spatial visualization.

Link: https://antarikchya.org.np/visualization

- ArcGIS Pro, QGIS, HEC-HMS, Google Earth Pro, Google Earth Engine (GEE), WebGIS, Php

Space-enabled Urban Solutions in Nepal (SUSN) *August 2023 – July 2024*

Satellite Data Analyst | GIS Analyst

- Played a key role in the Space-enabled Urban Solutions in Nepal (SUSN) project, collaborating with Kathmandu Metropolitan City (KMC) departments including CPC, DMD, Traffic Management and Road Safety, and EMD.
- Performed satellite data analysis and tailored Land Use/Land Cover (LULC) classification using geospatial technologies.
- Worked closely with cross-functional teams to understand departmental requirements and deliver GIS-based solutions.
- Developed GIS maps to support data-driven urban planning and disaster risk management decision-making.

Link: https://www.kmc.antarikchya.org.np

- ArcGIS Pro, QGIS, AutoCad, Google Earth Pro, Google Earth Engine (GEE), Python

Land Use/Land Cover Classification and Future Prediction *March 2024 – August 2024*

Remote Sensing Analyst | GIS Mapping

- Collaborated with Ranadheer Reddy at Asian Institute of Technology on LULC Classification and Future Prediction.
- Performed LULC mapping using Google Earth Engine and modeled future land use changes using TerrSet 2020.

- ArcGIS Pro, Google Earth Engine (GEE), TerrSet2020

Changunarayan Municipality *June 2023 – July 2023*

GIS Analyst

- Contributed to the digitization project in Changunarayan Municipality by manually digitizing buildings using high-resolution drone imagery.
- Performed road network analysis to determine road directions and improve spatial data accuracy for urban planning.
- Supported creation of accurate geospatial datasets for municipal-level planning and development activities.

Link: https://changu-stag.naxa.com.np

- ArcGIS Pro, QGIS

National Building Exposure *June 2023 – July 2023*

- Contributed to the National Building Exposure project as a building digitizer by manually extracting building footprints from high-resolution imagery.

- Applied manual digitization techniques to ensure spatial accuracy of building data.
- Supported development of a comprehensive building exposure database for risk assessment and analysis.

Link : https://exposure.ndrrma.gov.np

- ArcGIS

Impact on COR Station due to Gorkha Earthquake 2015 ***November 2022 – March 2023***

GIS Analyst

- Conducted final year Diploma project on geodetic deformation analysis using control point data from UNAVCO.
- Processed and analyzed control point datasets to study ground displacement.
- Visualized spatial changes using ArcGIS for deformation assessment.
- Investigated the impact of the 2015 Gorkha Earthquake on COR Station by identifying shifts in control points caused by seismic activity.

- ArcGIS, UNAVCO

Topographical and Cadastral Surveying | Jiri, Nepal ***November 23 – December 02, 2023***

- Participated in topographical and cadastral surveying in Jiri using DGPS for establishing control points.
- Performed field data collection using total stations and prisms for hydropower, road alignment, transmission line, bridge, and canal projects.
- Processed and analyzed survey data for spatial accuracy and mapping purposes.
- Visualized topographic and cadastral datasets using ArcGIS for engineering and planning applications.

- ArcGIS, Total Station (TS), Differential Global Positioning Systems (DGPSs), Prism, Measuring Tape

CONFERENCE AND PAPERS

SpaceCon 2026 | Pulchowk, Lalitpur ***April 12, 2024***

Link: https://spacecon.antarikchya.org.np/

SpaceCon 2025 | Sukedhara Heights, Kathmandu ***April 11, 2024***

"Monsoon Flood Impact Analysis of Nepal's most populous and Capital City, Kathmandu" and "Sapta Koshi Flood: Turning Risk into a valuable opportunity"

Innovation and Transformation in the Development Effort through Science, Technology and Governance | Kathmandu, Nepal ***May 17 – 19, 2024***

Link: https://www.mitrakunj.org.np/international-conference

NYC Scientists' Conference 2081 | Jawalakhel, Lalitpur ***June 19, 2024***

Link: https://www.ran.org.np

SpaceCon 2024 | Pulchowk, Lalitpur ***April 11, 2024***

PUBLICATIONS

Leveraging Remote Sensing for Lineament Extraction from Landsat-8, Sentinel-2 and Sentinel-1 at Bagmati Province of Nepal ***25 November 2025***

Link: https://ieeexplore.ieee.org/document/11242632

The Cubesat Case Study: Resource- Efficient Deep Learning for Space Application

Link: https://ieeexplore.ieee.org/document/11314016 ***25 November 2025***

TRAINING

Google Earth Engine - Hands-on skills in satellite data analysis and geospatial processing using the cloud-based Earth Engine platform.

Basic QGIS - How to visualize, analyze, and create maps using QGIS, including adding layers, styling maps, and performing basic spatial analysis like clip, buffer.

HEC-RAS & HEC-HMS - Performed rainfall-runoff modeling using HEC-HMS and simulated flood inundation maps using HEC-RAS for hydrologic and hydraulic analysis.

CERTIFICATIONS AND PARTICIPATIONS

- Cloud Native Remote Sensing with Python — ***June 23 – July 2, 2026***
- Python Foundation for Spatial Analysis — ***December 02 – 10, 2025***
- Google Earth Engine for Water Resources Management — ***August 04 – 20, 2025***
- Secured the 1st position in the Map Design Competition: https://www.facebook.com/share/p/1GKWoL6aWL — ***June 27, 2024***
- Honorable Mention for Compelling Visuals winner for NASA's Pale Blue Dot: Visualization Challenge: https://drivendata-public-assets.s3.amazonaws.com/nasa-open-science-gallery/nayan.bakhadyo/visual.png — ***March 14, 2024***
- Training on Google Earth Engine — ***January 18 – May 02, 2024***
- Training on PyQGIS Masterclass — ***July 15 – August 9, 2024***
- Training on Advanced 30DayOfQGIS — ***May 13 – June 11, 2024***
- Training on Land Use & GIS for Technical Personnel of Municipalities within Kathmandu Valley — ***January 29, 2024 – February 2, 2024***
- Training on Web GIS — ***December 2023 – February 2024***
- Esri MOOC — ***2023***
- Training on AutoCad — ***November 07 – 08, 2023***
- Training on Python — ***June 16 – 26 , 2023***
- Training on OpenStreetMap — ***August 17 – 18, 2022***

SKILLS

- Project management
- GIS (ArcGIS Pro, QGIS, ArcGIS)
- WebGIS
- Autocad
- Sketchup
- HEC-HMS | HEC-RAS
- ERDAS Imagine
- SNAP Desktop
- Microsoft Office
- TerrSet 2020
- Total Station
- Communication
- C
- C++
- Google Earth Engine
- ENVI
- Python
- HTML/CSS/JS
- PHP
- Theodolite
- Plane Table
- Differential Global Positioning Systems

WORK REFERENCES

- **Dr. Abhas Maskey**
  - Founder | Managing Director, Antarikchya Pratisthan Nepal
  - Email: abhas@antarikchya.org.np
  - Phone: +977 9869397707
- **Prof. Dr.-Ing. Ramesh Kumar Maskey**
  - Academician, Nepal Academy of Science and Technology (NAST)
  - Email: rmaskey@ku.edu.np
  - Phone: +977 9851102669
- **Er. Anukul Nidhi**
  - Assistant Traffic Engineer, City Planning Commission, Kathmandu Metropolitan City
  - Email: nidhianukul479@gmail.com
  - Phone: +977 9844417900
- **Er. Sujan Nepali**
  - Geospatial Consultant at ICIMOD | Geomatics Instructor at NBPI
  - Email: nepsujan17@gmail.com
  - Phone: +977 9861291592